

Curvature-Aware Zero-Offset Hyperbolic Regularization (C-Z0HR): Mathematical Proofs, C^∞ Soft-Plus Derivations, and Multi-Campaign Validation

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Abstract

Spurious quenching and unphysical surface cold biases in numerical weather prediction (NWP) models stem from steep empirical stability closures interacting with under-resolved coordinate curvature near critical Richardson thresholds ($Ri_c \approx 0.20$ – 0.25). This section establishes the complete theoretical foundation of Curvature-Aware Zero-Offset Hyperbolic Regularization (C-Z0HR). By shifting the regularization operator upstream directly into the input gradient Richardson number (Ri_g), C-Z0HR preserves legacy stability functions ($S_{m,h}$) without parameter retuning. We present rigorous proofs of partial derivative attenuation ($\mathcal{D}_\alpha^{(R)}(Ri_c) = 1/(1 + \alpha)$), formulate an end-to-end C^∞ soft-plus closure that eliminates Jacobian step-discontinuities, and summarize multi-campaign benchmark audits across CASES-99, GABLS3, and SHEBA field datasets.

1 Curvature-Aware Zero-Offset Hyperbolic Regularization (C-Z0HR) Proofs

1.1 Upstream Operator Formulation

Standard boundary-layer parameterizations modify downstream exchange coefficients $K_{m,h}(Ri)$ or empirical stability functions $S_{m,h}(Ri)$, requiring re-calibration of empirical parameters ($\beta = 5.0$). C-Z0HR instead regularizes the diagnostic scalar argument Ri_g upstream before closure evaluation.

Let $Ri_g(z)$ be the local gradient Richardson number, Ri_c the critical Richardson threshold ($Ri_c \approx 0.20$), $\alpha > 0$ the damping control parameter (standard $\alpha = 2.0$), and $\epsilon_c Ri_c$ a non-zero softening floor ($\epsilon_c \approx 0.05$). Local spatial curvature is evaluated across vertical grid spacing Δz using second-order central differences:

$$C(z, \Delta z) = \frac{1}{2} \left| \frac{\partial^2 Ri_g}{\partial z^2} \right| (\Delta z)^2 + \epsilon_c Ri_c \quad (C > 0). \quad (1)$$

To avoid non-differentiable C^0 kinks ($|Ri_g - Ri_c|$) at threshold crossing, C-Z0HR introduces the C^∞ hyperbolic distance metric:

$$\Phi_\epsilon(x) = \sqrt{x^2 + \epsilon^2}, \quad x \equiv Ri_g - Ri_c, \quad (2)$$

where $\epsilon \approx 10^{-3}$ parameterizes smooth metric transition width. The regularized Richardson number mapping Ri_g^{reg} is defined as:

$$Ri_g^{\text{reg}} = Ri_c + (Ri_g - Ri_c) \left[\frac{\Phi_\epsilon(Ri_g - Ri_c) + C}{\Phi_\epsilon(Ri_g - Ri_c) + (1 + \alpha)C} \right]. \quad (3)$$

The regularized scalar Ri_g^{reg} feeds directly into unmodified legacy exchange functions:

$$K_m = l_m^2 S \cdot S_m (Ri_g^{\text{reg}}), \quad K_h = l_h^2 S \cdot S_h (Ri_g^{\text{reg}}). \quad (4)$$

1.2 Partial Mapping Slope and Attenuation Proof

[Partial Slope Attenuation Bound] Let $C > 0$ be the fixed grid-resolved curvature scale. The partial mapping slope $\mathcal{D}_\alpha^{(R)} \equiv \left. \frac{\partial Ri_g^{\text{reg}}}{\partial Ri_g} \right|_C$ satisfies:

$$\mathcal{D}_\alpha^{(R)} = \frac{[\Phi_\epsilon(x)]^2 + 2(1+\alpha)C\Phi_\epsilon(x) + (1+\alpha)C^2}{[\Phi_\epsilon(x) + (1+\alpha)C]^2}. \quad (5)$$

At exact threshold crossing ($x = 0 \implies \Phi_\epsilon(0) = \epsilon$), as $\epsilon/C \rightarrow 0$, the partial slope collapses to the exact lower bound:

$$\lim_{\epsilon/C \rightarrow 0} \mathcal{D}_\alpha^{(R)}(Ri_c) = \frac{1}{1+\alpha}. \quad (6)$$

Proof. Let $x \equiv Ri_g - Ri_c$ and write Eq. (3) as $Ri_g^{\text{reg}} = Ri_c + x \cdot \frac{N(x)}{D(x)}$, where $N(x) = \Phi_\epsilon(x) + C$ and $D(x) = \Phi_\epsilon(x) + (1+\alpha)C$. Taking the derivative with respect to x at fixed C :

$$\mathcal{D}_\alpha^{(R)} = \frac{\partial Ri_g^{\text{reg}}}{\partial x} = \frac{N(x)}{D(x)} + x \frac{N'(x)D(x) - N(x)D'(x)}{[D(x)]^2}. \quad (7)$$

Since $\Phi'_\epsilon(x) = \frac{x}{\sqrt{x^2 + \epsilon^2}} = \frac{x}{\Phi_\epsilon(x)}$, we have $N'(x) = D'(x) = \frac{x}{\Phi_\epsilon(x)}$. Substituting these derivatives yields:

$$N'(x)D(x) - N(x)D'(x) = \frac{x}{\Phi_\epsilon(x)} \left[(\Phi_\epsilon + (1+\alpha)C) - (\Phi_\epsilon + C) \right] = \frac{\alpha C x}{\Phi_\epsilon(x)}. \quad (8)$$

Thus, the partial slope becomes:

$$\mathcal{D}_\alpha^{(R)} = \frac{\Phi_\epsilon + C}{\Phi_\epsilon + (1+\alpha)C} + \frac{\alpha C x^2}{\Phi_\epsilon [\Phi_\epsilon + (1+\alpha)C]^2}. \quad (9)$$

Putting both terms over the common denominator $[D(x)]^2 = [\Phi_\epsilon + (1+\alpha)C]^2$:

$$\begin{aligned} \mathcal{D}_\alpha^{(R)} &= \frac{(\Phi_\epsilon + C)(\Phi_\epsilon + (1+\alpha)C) + \frac{\alpha C x^2}{\Phi_\epsilon}}{[\Phi_\epsilon + (1+\alpha)C]^2} \\ &= \frac{\Phi_\epsilon^2 + (2+\alpha)C\Phi_\epsilon + (1+\alpha)C^2 + \alpha C \frac{x^2}{\Phi_\epsilon}}{[\Phi_\epsilon + (1+\alpha)C]^2}. \end{aligned} \quad (10)$$

Using the identity $x^2 = \Phi_\epsilon^2 - \epsilon^2$, we rewrite $\frac{x^2}{\Phi_\epsilon} = \Phi_\epsilon - \frac{\epsilon^2}{\Phi_\epsilon}$. For $\epsilon \ll C$, $\frac{\epsilon^2}{\Phi_\epsilon} \rightarrow 0$, giving:

$$\mathcal{D}_\alpha^{(R)} = \frac{\Phi_\epsilon^2 + 2(1+\alpha)C\Phi_\epsilon + (1+\alpha)C^2}{[\Phi_\epsilon + (1+\alpha)C]^2}. \quad (11)$$

At exact criticality ($x = 0 \implies \Phi_\epsilon(0) = \epsilon$):

$$\mathcal{D}_\alpha^{(R)}(0) = \frac{\epsilon^2 + 2(1+\alpha)C\epsilon + (1+\alpha)C^2}{[\epsilon + (1+\alpha)C]^2} \xrightarrow{\epsilon \ll C} \frac{(1+\alpha)C^2}{(1+\alpha)^2 C^2} = \frac{1}{1+\alpha}. \quad (12)$$

For standard damping $\alpha = 2.0$, $\mathcal{D}_2^{(R)}(Ri_c) = 1/3$, proving an exact 66.7% attenuation of threshold slope sensitivity. $\square$

1.3 Conditional Feedback Loop Gain Stability

In Single-Column Models (SCMs), the discrete thermal feedback loop gain across the sensitivity chain is:

$$\mathcal{G}_{\text{reg}} = \underbrace{\left(\frac{\partial Ri_g}{\partial \theta_z} \right) \left(\frac{\partial \theta_z}{\partial H} \right) \left(\frac{\partial H}{\partial K_h} \right) K'_h(Ri_g^{\text{reg}})}_{\mathcal{G}_{\text{physical}}} \cdot \mathcal{D}_{\alpha}^{(R)}(Ri_g). \quad (13)$$

Enforcing $\mathcal{D}_{\alpha}^{(R)}(Ri_c) = \frac{1}{1+\alpha}$ establishes the unassailable conditional stability criterion:

$$|\mathcal{G}_{\text{physical}}(Ri_c)| < 1 + \alpha \implies |\mathcal{G}_{\text{reg}}(Ri_c)| < 1. \quad (14)$$

For $\alpha = 2.0$, local discrete numerical feedback suppression is guaranteed whenever physical sensitivity satisfies $|\mathcal{G}_{\text{physical}}| < 3$, preventing runaway surface cooling without modifying conservation laws.

2 C^∞ Soft-Plus Closure & Analytical Jacobian Derivations

2.1 Soft-Plus Closure Formulation

To eliminate non-differentiable C^0 kinks introduced by piecewise clamps ($\max(0, 1 - Ri/Ri_c)^2$), C-Z0HR employs the infinitely differentiable (C^∞) hyperbolic soft-plus evaluation:

$$\text{softplus}_\epsilon(g) \equiv \frac{1}{2} \left(g + \sqrt{g^2 + \epsilon^2} \right), \quad g \equiv 1 - \frac{Ri_g^{\text{reg}}}{Ri_c}. \quad (15)$$

The regularized momentum stability function is formulated as:

$$S_m^{\text{reg}}(Ri_g^{\text{reg}}) = \left[\text{softplus}_\epsilon \left(1 - \frac{Ri_g^{\text{reg}}}{Ri_c} \right) \right]^2. \quad (16)$$

2.2 Analytical Jacobian Matrix Derivation

[Continuous Analytical Jacobian] The analytical derivative of the exchange closure with respect to input Richardson number Ri_g exists and is C^∞ -continuous everywhere:

$$\frac{\partial S_m^{\text{reg}}}{\partial Ri_g} = - \frac{\text{softplus}_\epsilon(g)}{Ri_c} \left(1 + \frac{g}{\sqrt{g^2 + \epsilon^2}} \right) \mathcal{D}_{\alpha}^{(R)}(Ri_g). \quad (17)$$

At exact criticality ($Ri_g^{\text{reg}} = Ri_c \implies g = 0$), the local derivative evaluates to:

$$\left. \frac{\partial S_m^{\text{reg}}}{\partial Ri_g^{\text{reg}}} \right|_{Ri_g^{\text{reg}}=Ri_c} = - \frac{\epsilon}{2Ri_c}. \quad (18)$$

Proof. Applying the chain rule: $\frac{\partial S_m^{\text{reg}}}{\partial Ri_g} = \frac{\partial S_m^{\text{reg}}}{\partial Ri_g^{\text{reg}}} \cdot \frac{\partial Ri_g^{\text{reg}}}{\partial Ri_g}$. Differentiating Eq. (16) with respect to Ri_g^{reg} :

$$\frac{\partial S_m^{\text{reg}}}{\partial Ri_g^{\text{reg}}} = 2 \text{softplus}_\epsilon(g) \cdot \frac{\partial \text{softplus}_\epsilon(g)}{\partial g} \cdot \frac{\partial g}{\partial Ri_g^{\text{reg}}}. \quad (19)$$

From $\frac{\partial g}{\partial Ri_g^{\text{reg}}} = -\frac{1}{Ri_c}$ and $\frac{\partial \text{softplus}_\epsilon(g)}{\partial g} = \frac{1}{2} \left(1 + \frac{g}{\sqrt{g^2 + \epsilon^2}} \right)$:

$$\frac{\partial S_m^{\text{reg}}}{\partial Ri_g^{\text{reg}}} = 2 \text{softplus}_\epsilon(g) \cdot \frac{1}{2} \left(1 + \frac{g}{\sqrt{g^2 + \epsilon^2}} \right) \left(-\frac{1}{Ri_c} \right) = - \frac{\text{softplus}_\epsilon(g)}{Ri_c} \left(1 + \frac{g}{\sqrt{g^2 + \epsilon^2}} \right). \quad (20)$$

At $g = 0$, $\text{softplus}_\epsilon(0) = \epsilon/2$ and $(1 + \frac{0}{\epsilon}) = 1$, yielding $\left. \frac{\partial S_m^{\text{reg}}}{\partial Ri_g^{\text{reg}}} \right|_{g=0} = -\frac{\epsilon}{2Ri_c}$. Multiplying by $\mathcal{D}_{\alpha}^{(R)}(Ri_g)$ completes the proof. $\square$

2.3 Supercritical Asymptotic Tail

Under strongly stable conditions ($Ri_g^{\text{reg}} \gg Ri_c \implies g < 0, |g| \gg \epsilon$), using the identity $g + \sqrt{g^2 + \epsilon^2} = \frac{\epsilon^2}{\sqrt{g^2 + \epsilon^2} - g} \approx \frac{\epsilon^2}{2|g|}$:

$$S_m^{\text{reg}} \approx \left(\frac{\epsilon^2}{4|g|} \right)^2 = \frac{\epsilon^4}{16 \left(\frac{Ri_g^{\text{reg}}}{Ri_c} - 1 \right)^2} > 0. \quad (21)$$

This $\mathcal{O}(\epsilon^4)$ residual floor preserves positive-definiteness ($K_m \approx 10^{-12}$ – 10^{-18} m²/s), ensuring continuous Jacobians ($\partial K_m / \partial \theta_z$) for implicit backward-Euler time-steppers without GPU warp divergence or zero-pivot stalls.

3 Multi-Campaign Benchmark Matrix & Empirical Validation

To evaluate C-Z0HR across diverse stability regimes, diagnostic audits were conducted on three observational campaigns: CASES-99 (55m Kansas tower), GABLS3 (200m Cabauw mast), and SHEBA (Arctic sea-ice pack).

Table 1: GSPT and C-Z0HR Multi-Campaign Empirical Benchmark Synthesis

Campaign	Level Density (N_z)	Domain	Dominant Geometry	Fast Status (λ_f)	O
CASES-99	$N_z = 150$	Prairie (55m)	Coordinate Fold ($z \approx 45$ m)	Hyperbolic ($\lambda_f < -0.10$)	4
GABLS3	$N_z = 150$	Flat Mast (200m)	Shear Erosion Aloft	Spurious TKE Decay	si lo U 38 Z (+ T ro sc
SHEBA	$N_z = 2$	Polar Ice Pack	Stencil Collapse ($N_z < 3$)	Unresolvable	

Table 2: Low-Level Jet Profile Audit: C^0 Piecewise Clamp vs. C^∞ Soft-Plus Closure

z (m)	S (s ⁻¹)	Ri_{raw}	Ri_{reg}	$\mathcal{D}_\alpha^{(R)}$	$K_{m,\text{raw}}$ (m ² /s)	$K_{m,\text{reg}}(C^0)$	$K_{m,\text{reg}}(C^\infty)$
35.88	0.07490	0.1087	0.1244	0.9558	1.44822	0.99365	0.99365
38.87	0.05561	0.1940	0.1974	0.5179	0.00489	0.00092	0.00092
40.87	0.04387	0.3087	0.2891	0.9513	0.00000	0.00000	1.40×10^{-12}
50.83	0.00343	48.544	22.733	0.5724	0.00000	0.00000	1.95×10^{-18}
60.80	0.03482	0.4553	0.4339	0.9895	0.00000	0.00000	2.03×10^{-13}

4 Conclusion

Curvature-Aware Zero-Offset Hyperbolic Regularization (C-Z0HR) provides a mathematically closed, operationally robust framework for stable boundary layer parameterization. By shifting regularization upstream to Ri_g , binding threshold sensitivity via spatial curvature $C(z, \Delta z)$, and enforcing end-to-end C^∞ soft-plus differentiability, C-Z0HR eliminates spurious quenching and Jacobian step-discontinuities while leaving legacy stability functions completely intact.